Tooling Validation Report - New Injection Mold - Section 1: Executive Summary & Scope

Validation results for the newly commissioned injection mold tool, covering first article inspection and short-run capability. Section 1 (Executive Summary & Scope) applies to Process Engineering within Manufacturing Knowledge Hub and is maintained by Carlos Ibarra. Defines the operational charter, scope boundaries, and executive accountability.

Operational requirement 1.1: First article inspection passed all 14 critical dimensions within tolerance. Secondary control 1.2: Short-run capability study produced a Cpk of 1.48 on the primary wall thickness dimension.

- First article inspection passed all 14 critical dimensions within tolerance.
- Short-run capability study produced a Cpk of 1.48 on the primary wall thickness dimension.
- Section 1 review cadence is tracked by Carlos Ibarra under Process Engineering governance.

Control Area	Owner	Target SLA	Verification Status
Process Engineering	Carlos Ibarra	5 business days	Verified
Tooling Validation Report - Gate 1	Carlos Ibarra	3 business days	Active
Tooling Validation Report - Gate 2	Carlos Ibarra	10 business days	Monitored

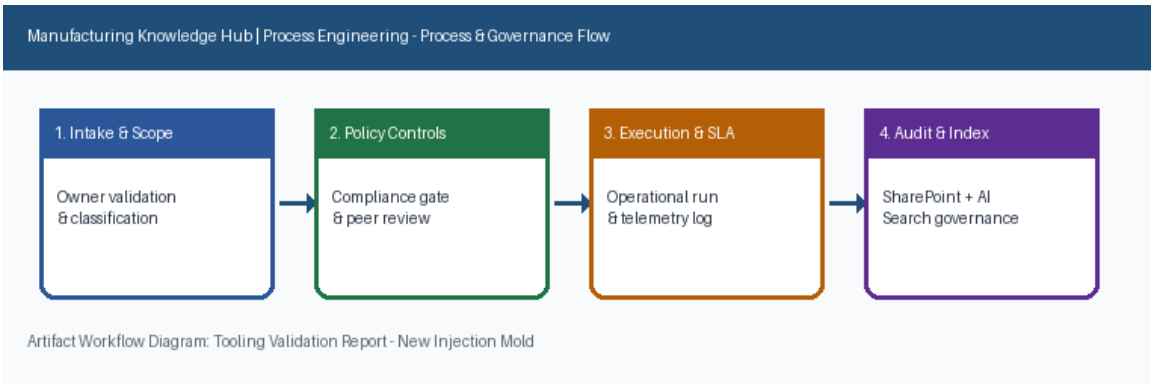
Tooling Validation Report - New Injection Mold - Section 2: Architecture & Process Flow

Validation results for the newly commissioned injection mold tool, covering first article inspection and short-run capability. Section 2 (Architecture & Process Flow) applies to Process Engineering within Manufacturing Knowledge Hub and is maintained by Carlos Ibarra. Maps end-to-end process stages, handoff gates, and system-of-record integrations.

Operational requirement 2.1: Short-run capability study produced a Cpk of 1.48 on the primary wall thickness dimension. Secondary control 2.2: Mold flow simulation matched actual fill pattern within acceptable variance.

- Short-run capability study produced a Cpk of 1.48 on the primary wall thickness dimension.
- Mold flow simulation matched actual fill pattern within acceptable variance.
- Section 2 review cadence is tracked by Carlos Ibarra under Process Engineering governance.

Control Area	Owner	Target SLA	Verification Status
Process Engineering	Carlos Ibarra	5 business days	Verified
Tooling Validation Report - Gate 1	Carlos Ibarra	3 business days	Active
Tooling Validation Report - Gate 2	Carlos Ibarra	10 business days	Monitored
Sec-2	Carlos Ibarra	3 business days	Stage 2 Verified



Source: Manufacturing Knowledge Hub knowledge base | SharePoint Online

Tooling Validation Report - New Injection Mold - Section 3: Governance Standards & Controls

Validation results for the newly commissioned injection mold tool, covering first article inspection and short-run capability. Section 3 (Governance Standards & Controls) applies to Process Engineering within Manufacturing Knowledge Hub and is maintained by Carlos Ibarra. Establishes mandatory compliance controls, approval thresholds, and audit evidence.

Operational requirement 3.1: Mold flow simulation matched actual fill pattern within acceptable variance. Secondary control 3.2: Tool is approved for production release pending preventive maintenance plan sign-off.

- Mold flow simulation matched actual fill pattern within acceptable variance.
- Tool is approved for production release pending preventive maintenance plan sign-off.
- Section 3 review cadence is tracked by Carlos Ibarra under Process Engineering governance.

Control Area	Owner	Target SLA	Verification Status
Process Engineering	Carlos Ibarra	5 business days	Verified
Tooling Validation Report - Gate 1	Carlos Ibarra	3 business days	Active
Tooling Validation Report - Gate 2	Carlos Ibarra	10 business days	Monitored
Sec-3	Carlos Ibarra	10 business days	Stage 3 Verified

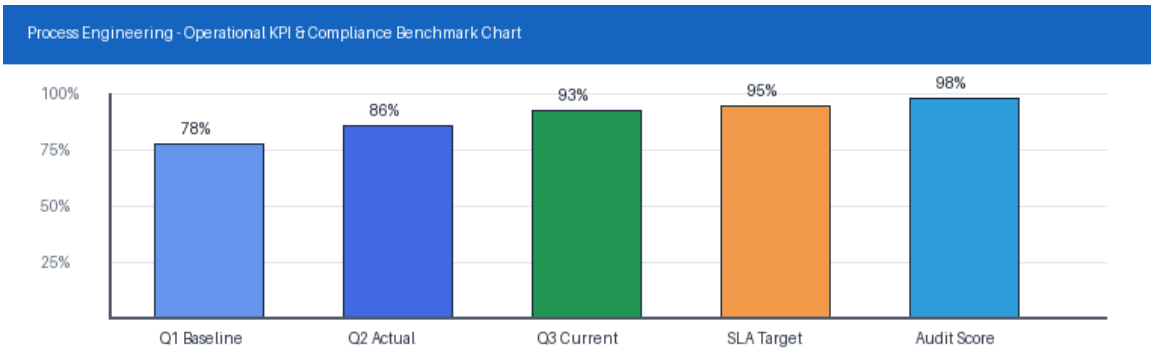
Tooling Validation Report - New Injection Mold - Section 4: Operational KPIs & Benchmarks

Validation results for the newly commissioned injection mold tool, covering first article inspection and short-run capability. Section 4 (Operational KPIs & Benchmarks) applies to Process Engineering within Manufacturing Knowledge Hub and is maintained by Carlos Ibarra. Tracks baseline performance, target SLAs, and quarterly benchmark progression.

Operational requirement 4.1: Tool is approved for production release pending preventive maintenance plan sign-off. Secondary control 4.2: First article inspection passed all 14 critical dimensions within tolerance.

- Tool is approved for production release pending preventive maintenance plan sign-off.
- First article inspection passed all 14 critical dimensions within tolerance.
- Section 4 review cadence is tracked by Carlos Ibarra under Process Engineering governance.

Control Area	Owner	Target SLA	Verification Status
Process Engineering	Carlos Ibarra	5 business days	Verified
Tooling Validation Report - Gate 1	Carlos Ibarra	3 business days	Active
Tooling Validation Report - Gate 2	Carlos Ibarra	10 business days	Monitored
Sec-4	Carlos Ibarra	5 business days	Stage 4 Verified



Source: Manufacturing Knowledge Hub knowledge base | SharePoint Online

Tooling Validation Report - New Injection Mold - Section 5: Roles, Responsibilities & RACI

Validation results for the newly commissioned injection mold tool, covering first article inspection and short-run capability. Section 5 (Roles, Responsibilities & RACI) applies to Process Engineering within Manufacturing Knowledge Hub and is maintained by Carlos Ibarra. Clarifies ownership across accountable executives, operational leads, and reviewers.

Operational requirement 5.1: First article inspection passed all 14 critical dimensions within tolerance. Secondary control 5.2: Short-run capability study produced a Cpk of 1.48 on the primary wall thickness dimension.

- First article inspection passed all 14 critical dimensions within tolerance.
- Short-run capability study produced a Cpk of 1.48 on the primary wall thickness dimension.
- Section 5 review cadence is tracked by Carlos Ibarra under Process Engineering governance.

Control Area	Owner	Target SLA	Verification Status
Process Engineering	Carlos Ibarra	5 business days	Verified
Tooling Validation Report - Gate 1	Carlos Ibarra	3 business days	Active
Tooling Validation Report - Gate 2	Carlos Ibarra	10 business days	Monitored
Sec-5	Carlos Ibarra	3 business days	Stage 5 Verified

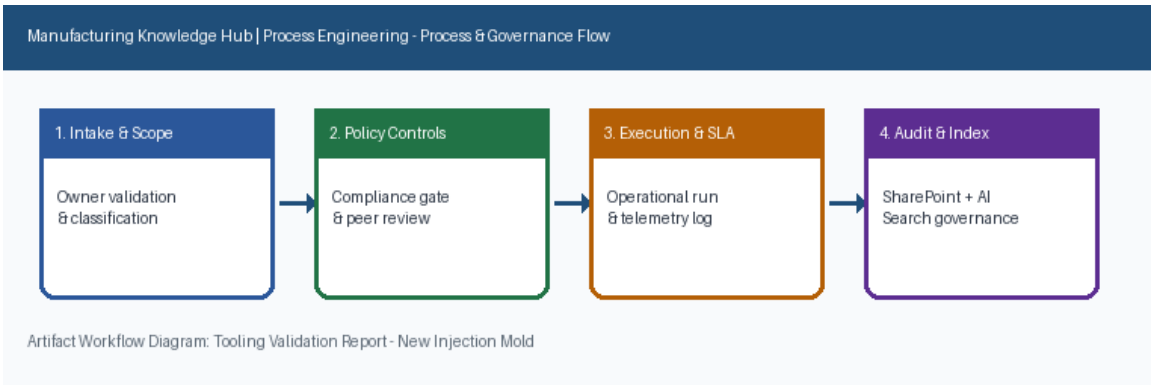
Tooling Validation Report - New Injection Mold - Section 6: Standard Operating Procedures

Validation results for the newly commissioned injection mold tool, covering first article inspection and short-run capability. Section 6 (Standard Operating Procedures) applies to Process Engineering within Manufacturing Knowledge Hub and is maintained by Carlos Ibarra. Provides step-by-step execution checklists, intake criteria, and verification gates.

Operational requirement 6.1: Short-run capability study produced a Cpk of 1.48 on the primary wall thickness dimension. Secondary control 6.2: Mold flow simulation matched actual fill pattern within acceptable variance.

- Short-run capability study produced a Cpk of 1.48 on the primary wall thickness dimension.
- Mold flow simulation matched actual fill pattern within acceptable variance.
- Section 6 review cadence is tracked by Carlos Ibarra under Process Engineering governance.

Control Area	Owner	Target SLA	Verification Status
Process Engineering	Carlos Ibarra	5 business days	Verified
Tooling Validation Report - Gate 1	Carlos Ibarra	3 business days	Active
Tooling Validation Report - Gate 2	Carlos Ibarra	10 business days	Monitored
Sec-6	Carlos Ibarra	10 business days	Stage 6 Verified



Source: Manufacturing Knowledge Hub knowledge base | SharePoint Online

Tooling Validation Report - New Injection Mold - Section 7: Risk Register & Mitigations

Validation results for the newly commissioned injection mold tool, covering first article inspection and short-run capability. Section 7 (Risk Register & Mitigations) applies to Process Engineering within Manufacturing Knowledge Hub and is maintained by Carlos Ibarra. Catalogs operational, compliance, and delivery risks alongside preventive mitigations.

Operational requirement 7.1: Mold flow simulation matched actual fill pattern within acceptable variance. Secondary control 7.2: Tool is approved for production release pending preventive maintenance plan sign-off.

- Mold flow simulation matched actual fill pattern within acceptable variance.
- Tool is approved for production release pending preventive maintenance plan sign-off.
- Section 7 review cadence is tracked by Carlos Ibarra under Process Engineering governance.

Control Area	Owner	Target SLA	Verification Status
Process Engineering	Carlos Ibarra	5 business days	Verified
Tooling Validation Report - Gate 1	Carlos Ibarra	3 business days	Active
Tooling Validation Report - Gate 2	Carlos Ibarra	10 business days	Monitored
Sec-7	Carlos Ibarra	5 business days	Stage 7 Verified

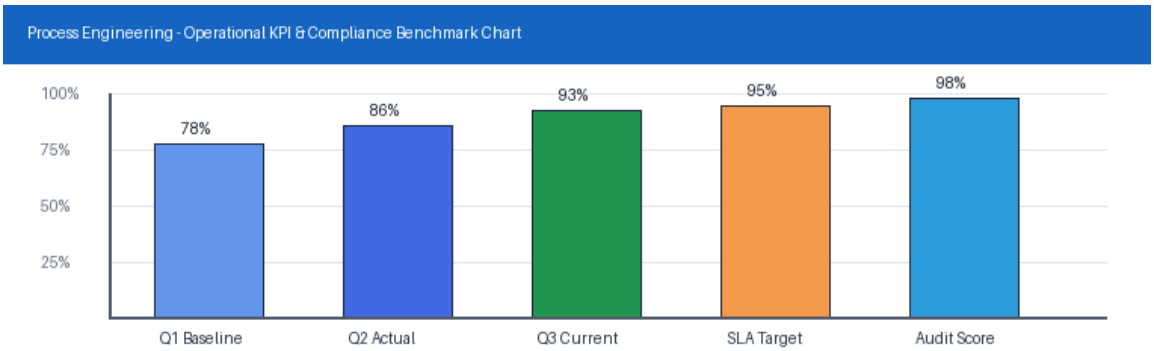
Tooling Validation Report - New Injection Mold - Section 8: Quality Assurance & Audits

Validation results for the newly commissioned injection mold tool, covering first article inspection and short-run capability. Section 8 (Quality Assurance & Audits) applies to Process Engineering within Manufacturing Knowledge Hub and is maintained by Carlos Ibarra. Specifies sampling frequencies, inspection tolerances, and nonconformance workflows.

Operational requirement 8.1: Tool is approved for production release pending preventive maintenance plan sign-off. Secondary control 8.2: First article inspection passed all 14 critical dimensions within tolerance.

- Tool is approved for production release pending preventive maintenance plan sign-off.
- First article inspection passed all 14 critical dimensions within tolerance.
- Section 8 review cadence is tracked by Carlos Ibarra under Process Engineering governance.

Control Area	Owner	Target SLA	Verification Status
Process Engineering	Carlos Ibarra	5 business days	Verified
Tooling Validation Report - Gate 1	Carlos Ibarra	3 business days	Active
Tooling Validation Report - Gate 2	Carlos Ibarra	10 business days	Monitored
Sec-8	Carlos Ibarra	3 business days	Stage 8 Verified



Source: Manufacturing Knowledge Hub knowledge base | SharePoint Online

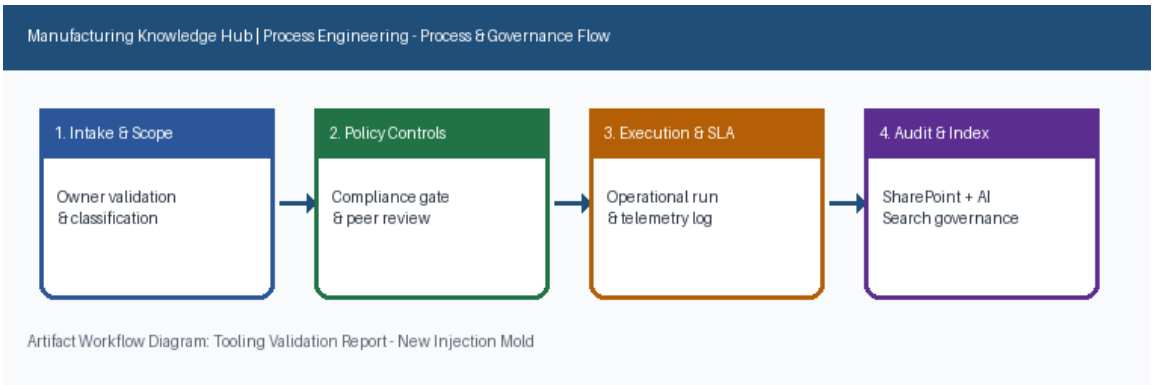
Tooling Validation Report - New Injection Mold - Section 9: Exception Handling & Escalations

Validation results for the newly commissioned injection mold tool, covering first article inspection and short-run capability. Section 9 (Exception Handling & Escalations) applies to Process Engineering within Manufacturing Knowledge Hub and is maintained by Carlos Ibarra. Outlines severity tiers, response SLAs, and cross-functional escalation paths.

Operational requirement 9.1: First article inspection passed all 14 critical dimensions within tolerance. Secondary control 9.2: Short-run capability study produced a Cpk of 1.48 on the primary wall thickness dimension.

- First article inspection passed all 14 critical dimensions within tolerance.
- Short-run capability study produced a Cpk of 1.48 on the primary wall thickness dimension.
- Section 9 review cadence is tracked by Carlos Ibarra under Process Engineering governance.

Control Area	Owner	Target SLA	Verification Status
Process Engineering	Carlos Ibarra	5 business days	Verified
Tooling Validation Report - Gate 1	Carlos Ibarra	3 business days	Active
Tooling Validation Report - Gate 2	Carlos Ibarra	10 business days	Monitored
Sec-9	Carlos Ibarra	10 business days	Stage 9 Verified



Source: Manufacturing Knowledge Hub knowledge base | SharePoint Online

Tooling Validation Report - New Injection Mold - Section 10: Cross-Functional Dependencies

Validation results for the newly commissioned injection mold tool, covering first article inspection and short-run capability. Section 10 (Cross-Functional Dependencies) applies to Process Engineering within Manufacturing Knowledge Hub and is maintained by Carlos Ibarra. Documents upstream and downstream team dependencies and service-level commitments.

Operational requirement 10.1: Short-run capability study produced a Cpk of 1.48 on the primary wall thickness dimension. Secondary control 10.2: Mold flow simulation matched actual fill pattern within acceptable variance.

- Short-run capability study produced a Cpk of 1.48 on the primary wall thickness dimension.
- Mold flow simulation matched actual fill pattern within acceptable variance.
- Section 10 review cadence is tracked by Carlos Ibarra under Process Engineering governance.

Control Area	Owner	Target SLA	Verification Status
Process Engineering	Carlos Ibarra	5 business days	Verified
Tooling Validation Report - Gate 1	Carlos Ibarra	3 business days	Active
Tooling Validation Report - Gate 2	Carlos Ibarra	10 business days	Monitored
Sec-10	Carlos Ibarra	5 business days	Stage 10 Verified

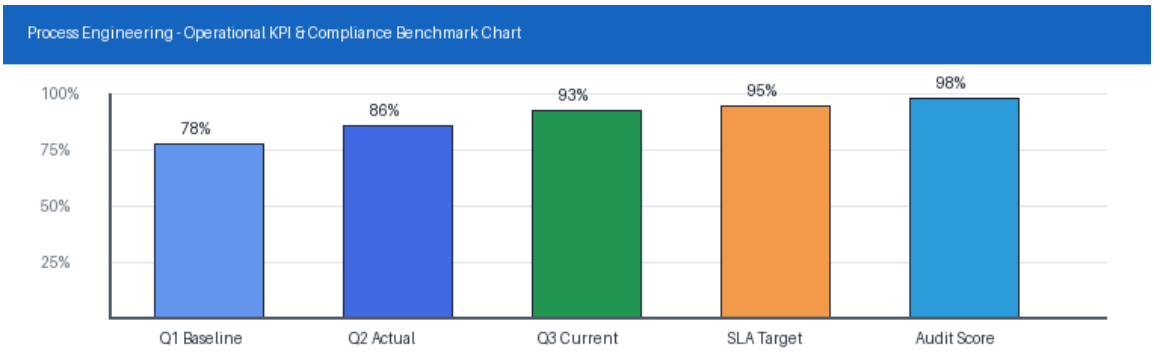
Tooling Validation Report - New Injection Mold - Section 11: Appendix, Lineage & Sign-off

Validation results for the newly commissioned injection mold tool, covering first article inspection and short-run capability. Section 11 (Appendix, Lineage & Sign-off) applies to Process Engineering within Manufacturing Knowledge Hub and is maintained by Carlos Ibarra. Records document lineage, governance metadata, version history, and owner sign-off.

Operational requirement 11.1: Mold flow simulation matched actual fill pattern within acceptable variance. Secondary control 11.2: Tool is approved for production release pending preventive maintenance plan sign-off.

- Mold flow simulation matched actual fill pattern within acceptable variance.
- Tool is approved for production release pending preventive maintenance plan sign-off.
- Section 11 review cadence is tracked by Carlos Ibarra under Process Engineering governance.

Control Area	Owner	Target SLA	Verification Status
Process Engineering	Carlos Ibarra	5 business days	Verified
Tooling Validation Report - Gate 1	Carlos Ibarra	3 business days	Active
Tooling Validation Report - Gate 2	Carlos Ibarra	10 business days	Monitored
Sec-11	Carlos Ibarra	3 business days	Stage 11 Verified



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